SUSTAINABLE INDUSTRY SPRINT

Type : Group (4 members)

Duration : 3.5 - 4hours

“Design the Future. Solve for Sustainability.”

EVENT OBJECTIVES

- **Expose students to real industrial sustainability challenges.**
- **Develop structured problem-solving under time constraints.**
- **Improve professional communication and decision-making.**
- **Encourage interdisciplinary collaboration and strengthen presentation.**

⏳ EVENT FLOW & TIMELINE

🟢 **0:00 – 0:20 | Registration & Orientation** What

Happens: • Team verification • Distribution of A4

problem analysis sheets • Rules & evaluation criteria

explained

• Judges introduced (Faculty / Alumni / Industry Guest).

🟢 **0:20 – 0:30 | Problem Statement Reveal**

• One problem sheet is revealed to all teams.

• All teams receive identical data sheets.

• No prior preparation allowed.

Teams must work within given:

- Budget constraints
- Timeline limits
- Technical restrictions

🟢 **0:30 – 2:00 | Analysis Phase (90 Minutes)**

Teams must:

✓ Clearly define the problem

✓ Identify root causes

✓ Brainstorm 2–3 solutions

✓ Select the best solution

✓ Estimate cost

✓ Analyze sustainability impact

✓ Assess risks and feasibility

✓ Develop implementation timeline

Participants record their structured thinking on provided A4 sheets.

🟢 **2:00 – 3:00 | Presentation Preparation (60 Minutes)**

Teams prepare a 5–7 slide PPT including:

- Problem Summary
- Key Insight
- Proposed Sustainable Solution
- Cost Breakdown
- Environmental Impact
- Implementation Plan
- Why This Works

🟢 **3:00 – 3:50 | Boardroom Presentation Round**

Each Team Gets: • 7–8 minutes Presentation • 5 minutes Q&A by Judges

Professional attire encouraged.

📜 RULES & REGULATIONS

1□ **Team Structure**

• Exactly 4 members per team

• Interdisciplinary teams encouraged

2 Event Format Rules

🔹 Problem Statement

- Revealed only at event start
- No external prior preparation allowed
- Same dataset provided to all teams

🔹 Time Limits

- **90 min –** Analysis
- **60 min –** PPT Preparation
- **7–8 min –** Final Presentation
- **5 min –** Q&A

🔹 Permitted Resources

- Laptops allowed
- Internet usage: (Decide – Allowed / Restricted / Not Allowed)

🔹 Mandatory Solution Components

Each team must include:

- Clear Problem Definition
- Root Cause Analysis
- Sustainable Solution Proposal
- Cost Estimation
- Environmental Impact Assessment
- Implementation Timeline
- Risk & Feasibility Analysis

📄 SAMPLE PROBLEM STATEMENTS

PROBLEM SHEET 1

EV Battery Waste Optimization – Clean Mobility Startup

🏢 **Company Background**

“VoltRide Mobility” is a mid-sized Indian EV startup manufacturing electric scooters.

They produce 25,000 lithium-ion battery packs per year.

⚠️ **Current Issue**

• 10% of batteries fail quality tests.

• Disposal cost per rejected battery: ₹2,200

• Increasing regulatory pressure on battery waste.

• Customers demanding sustainability transparency.

💰 **Constraints**

• Budget limit: ₹75 lakhs

• Cannot change battery supplier immediately.

• Solution must be implementable within 12 months.

🎯 **Target Outcome**

• Reduce battery waste by at least 50%.

• Lower disposal cost.

• Improve sustainability reputation.

PROBLEM SHEET 2

Sustainable Packaging Redesign – Food Manufacturing Company

🏭 **Company Background**

“FreshBite Foods” produces packaged snacks sold nationwide.

Annual packaging usage:

• 1.5 million plastic packets

• Cost per packet: ₹4.20

⚠️ **Current Issue**

• Plastic waste criticism from consumers.

• Retailers asking for eco-friendly packaging.

• Packaging must remain moisture-resistant.

💰 **Constraints**

• New packaging cost must stay under ₹5 per unit.

• Shelf life must remain 6 months minimum.

• Machines cannot be completely replaced.

🎯 **Target Outcome**

• Reduce plastic usage by at least 60%.

• Maintain product quality.

• Keep overall cost increase under 15%.

PROBLEM SHEET 3

Campus Energy Reduction Strategy – Private Engineering College.

🏭 **Background**

A 3,000-student engineering college spends **:**

• ₹1.2 crore annually on electricity.

• 40% of usage comes from classrooms and labs.

⚠️ **Current Issue**

• Rising electricity tariff (increase expected 20% next year).

• ACs and lights often left on unnecessarily.

• Administration wants visible sustainability initiative.

💰 **Constraints**

• Budget: ₹50 lakhs.

• Solar installation possible but limited rooftop space.

• Cannot disrupt academic schedule.

🎯 **Target Outcome**

• Reduce electricity bill by 25% within 18 months.

• Improve campus sustainability ranking.

• Achieve measurable carbon reduction.

REQUIREMENTS

📄 **Materials**

• Printed A4 Problem Sheets.

• Analysis Recording Sheets.

• A room with a projector- In order for the participants to explain/showcase their solution.

👩‍💻 **Participants**

• Must bring laptops.

• Formal or semi-formal attire recommended.

🎖️ **Rewards**

Prizes for the winning teams.
🌿 Sustainability-themed badges/stickers.

✨ PRIZE POOL : 10K